

CO-2 ASSESSMENT TOOL 1

COURSE CODE: CSA16 COURSE NAME: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre-processing techniques like Data cleaning, Data integration, Data transformation and data Reduction, for effective data preparation (BL3,BL4)

Analytical Problem Solving

1. Covariance Analysis in Health Insurance (10 Marks)

The table below shows the **annual premium increase rate (xi)** and the **number of claims per 100 policyholders (yi)** in a health insurance company.

Using the covariance formula, determine whether premium increase and claim frequency have a **positive or inverse relationship**.

Before computing covariance, calculate the **mean of x and y**.

Premium Increase % (xi)	Claims per 100 Policyholders (yi)
3.2	15
4.0	18
5.5	25
4.8	20

2. Data Smoothing using Equal-Frequency Binning (Healthcare) (5 Marks)

A hospital is analysing **patient blood pressure readings (systolic)** to identify trends. Due to minor fluctuations, the analyst performs **local smoothing using equal-frequency (equi-depth) binning**.

Sample Data Set (Systolic BP readings):

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

3. Equal-Width Binning for Risk Categorization (5 Marks)

An insurance company evaluates **Body Mass Index (BMI)** of 18 clients for risk classification.

To simplify analysis, they smooth the data by replacing values with **closest bin boundaries using equal-width partitioning**.

Sample Data Set (Sorted BMI):

18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

4. Data Reduction using Bin Medians (Medical Supply Costs) (5 Marks)

A hospital administration is reviewing the **cost of medical consumables (in ₹)**.

To reduce data for mining purposes, they discretize the sorted price records using **bin medians**.

Sample Data Set (Costs in ₹):

120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 180, 190, 200, 210, 230, 300

5. Standardization of Patient Age Data (5 Marks)

Given the following **ages of insured individuals**:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Standardize the variable by performing the following:

- Compute the **mean absolute deviation (MAD)** of age.
- Compute the **z-score for the first four observations**.

Answers:

1)

Premium Increase (x)	Claims (y)
3.2	15
4.0	18
5.5	25
4.8	20

Step 1: Calculate Mean

$$\bar{x} = \frac{3.2 + 4.0 + 5.5 + 4.8}{4} = 4.375$$

$$\bar{y} = \frac{15 + 18 + 25 + 20}{4} = 19.5$$

Step 2: Compute Deviations

x	y	(x- $\bar{x}$)	(y- $\bar{y}$)	Product
3.2	15	-1.175	-4.5	5.2875

x	y	(x- $\bar{x}$)	(y- $\bar{y}$)	Product
4.0	18	-0.375	-1.5	0.5625
5.5	25	1.125	5.5	6.1875
4.8	20	0.425	0.5	0.2125

$$\sum(x-\bar{x})(y-\bar{y})=12.25$$

Step 3: Covariance

$$\begin{aligned}\text{Cov}(x, y) &= \frac{\sum(x - \bar{x})(y - \bar{y})}{n} \\ &= \frac{12.25}{4} = 3.0625\end{aligned}$$

So, Mean of Premium Increase = **4.375%**, Mean of Claims = **19.5**, and **Covariance = 3.0625**

2) Given Data:

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

There are **27 values**.

Divide into **3 equal-frequency bins**

Bin 1

110, 112, 115, 115, 118, 120, 120, 122, 124

Mean

$$= \frac{1056}{9} = 117.33$$

Smoothed Bin

117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33

Bin 2

124, 126, 126, 126, 126, 130, 135, 135, 138

Mean

$$= \frac{1166}{9} = 129.56$$

Smoothed Bin

129.56 repeated 9 times.

Bin 3

138, 138, 140, 145, 150, 152, 160, 180, 138

(Equivalent sorted bin: 138, 138, 138, 140, 145, 150, 152, 160, 180)

Mean

$$= \frac{1341}{9} = 149.00$$

Smoothed Bin

149 repeated 9 times.

Final Smoothed Data

117.33 (×9), 129.56 (×9), 149 (×9)

3) Given Data

18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

Step 1: Calculate Width

Minimum = **18.5**

Maximum = **39.5**

Range

$$= 39.5 - 18.5 = 21$$

Using **3 equal-width bins**

Width

$$= \frac{21}{3} = 7$$

Bins

Bin 1: 18.5–25.5

Boundary values = 18.5 and 25.5

Smoothed Values

18.5, 18.5, 18.5, 25.5, 25.5

Bin 2: 25.5–32.5

Boundary values = 25.5 and 32.5

Smoothed Values

25.5, 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5

Bin 3: 32.5–39.5

Boundary values = 32.5 and 39.5

Smoothed Values

32.5, 32.5, 32.5, 32.5, 39.5

4) Step 1: Sort Data

120, 140, 145, 145, 148, 150, 150, 150, 155, 155, 155, 160, 160, 165, 170, 170, 175, 175, 175, 180, 190, 200, 210, 230, 300

Divide into **3 bins (9 values each)**.

Bin 1

120, 140, 145, 145, **148**, 150, 150, 150, 155

Median = **148**

Smoothed Bin

148 repeated 9 times.

Bin 2

155, 155, 155, 160, **160**, 165, 170, 170, 175

Median = **160**

Smoothed Bin

160 repeated 9 times.

Bin 3

175, 175, 175, 180, **190**, 200, 210, 230, 300

Median = **190**

Smoothed Bin

190 repeated 9 times.

5) Given Ages

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

(a) Mean

$$\bar{x} = \frac{330}{10} = 33$$

Mean Absolute Deviation (MAD)

Age	Deviation
18	15
22	11
25	8
42	9
28	5
43	10
33	0
35	2
56	23
28	5

Total deviation

$$= 88$$

$$MAD = \frac{88}{10} = 8.8$$

MAD = 8.8

(b) Z-score

Standard deviation

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{10}} = 10.835$$

Formula

$$z = \frac{x - \bar{x}}{\sigma}$$

Age	Z-score
18	-1.384
22	-1.015
25	-0.738
42	0.831